

ENTROPIA-Core: Dynamical Stability of the DME Vacuum

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Abstract

We derive ENTROPIA Core, the dynamical stability sector of the Dimensional Measurement Equilibrium (DME) vacuum, from DME plus the Entropic Damping Force (EDF). DME establishes that the vacuum entropy

$$S(\epsilon)/S_0 = 1 - \gamma\epsilon^2 + O(\epsilon^4), \quad \gamma = 0.70283946799007245929257819650854315,$$

is a strict local maximum at $\epsilon \equiv D - 4 = 0$. EDF derives, from an Ohmic hidden bath in the overdamped limit, a linear dissipative relaxation in the coarse-graining parameter. Identifying the EDF collective coordinate with the DME dimensional deviation closes the previously assumed gradient flow. The result is a Lyapunov functional $L[\epsilon] = \gamma\epsilon^2 + O(\epsilon^4)$, a gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$, monotone Lyapunov decrease $dL/d\lambda = -4\gamma^2\epsilon^2 \leq 0$, exponential relaxation $\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$, and relaxation time $\tau = 1/(2\gamma) = 0.711400\dots$. The Master conditions M1 and M2 follow as consequences. The identification constrains the hidden bath ratio T_b/γ_f to be of order unity, consistent with EDF's cold bath assumption; this is a prediction, not a fine tuning. No framework specific postulate is required beyond the standard Ohmic bath and overdamped assumptions of non-equilibrium statistical mechanics. All numerical content is inherited from DME's γ and is exact.

Contents

1	Introduction	2
2	Inputs from DME	2
3	The Lyapunov functional	4
4	EDF relaxation	4
5	The thermodynamic identification	5
6	The ENTROPIA-Core system	6
7	Master conditions	7
8	Explicit solution and relaxation time	8
9	Assumption audit	8
10	Relation to DME and ENTROPIA-EFT	8

11 Conclusion	9
A DME expansion coefficients	10
B EDF derivation summary	10
C Consistency between the noise rescaling and Proposition 7	11

1 Introduction

The Dimensional Measurement Equilibrium (DME) framework identifies the spacetime dimension D as a dynamical thermodynamic variable whose equilibrium value $D = 4$ is selected by maximization of holographic entanglement entropy [1]. This is a statement about the *static* equilibrium of the vacuum: $D = 4$ is a strict local maximum of the normalized entropy $S(\epsilon)/S_0$ where $\epsilon \equiv D - 4$.

A static maximum is not the same as a dynamical attractor. A system can sit at a maximum of a thermodynamic potential and still be dynamically unstable if the relaxation dynamics drive it away. ENTROPIA-Core addresses the dynamical question: given the DME entropy well, do perturbations of ϵ decay back to $\epsilon = 0$, and at what rate?

This paper derives the ENTROPIA Core sector end to end. The result is:

- A Lyapunov functional $L[\epsilon] = \gamma\epsilon^2 + O(\epsilon^4)$ (algebraic, from DME).
- A gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$ (from EDF, in the overdamped limit).
- A monotone Lyapunov decrease $dL/d\lambda = -4\gamma^2\epsilon^2 \leq 0$ (algebraic, given the flow).
- Master conditions M1 and M2 as consequences (definitions of equilibrium and monotone export).
- An explicit solution $\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$ and relaxation time $\tau = 1/(2\gamma) = 0.711400\dots$

The only place where an *identification* enters is the matching of the EDF collective coordinate σ (the fractional deviation of the effective coupling) with the DME dimensional deviation ϵ . These are two descriptions of the same relaxation mode, related by a proportionality that can be absorbed into the normalization of ϵ . We state this identification explicitly in §5, and we show there that it constrains the hidden-bath ratio T_b/γ_f to be of order unity — a prediction, not a fine-tuning.

The paper is organized as follows. Section 2 summarizes the DME inputs. Section 3 defines the Lyapunov functional. Section 4 summarizes the EDF derivation of linear damping. Section 5 performs the thermodynamic identification that closes the gradient flow. Section 6 derives the ENTROPIA-Core system. Section 7 states M1 and M2. Section 8 gives the explicit solution. Section 9 contains the assumption audit. Section 10 relates Core to DME and to ENTROPIA-EFT.

2 Inputs from DME

We take the following results from DME [1] as given. They are derived there from the regulated holographic strip integral and are exact.

Dimensional setup

With $D = 4$,

$$d = D - 1 = 3, \quad \alpha = D - 2 = 2.$$

The strip integral

The finite part of the holographic entanglement entropy for a strip is

$$I(\alpha) = \int_1^\infty du \left[\frac{u^{2\alpha-2}}{\sqrt{u^{2\alpha}-1}} - u^{\alpha-2} \right]. \quad (1)$$

This has the closed form

$$I(\alpha) = \frac{1}{2\alpha} \left[B\left(a, \frac{1}{2}\right) - \frac{1}{a} \right], \quad a = -\frac{1}{2} + \frac{1}{2\alpha}. \quad (2)$$

Exact numerical values

Evaluated to 50-digit precision:

$$I(2) = 0.40092988263220389628003875385983806, \quad (3)$$

$$I'(2) = -0.16567559766872346592538818546282266, \quad (4)$$

$$I''(2) = +0.1408946727052701891991682833342319. \quad (5)$$

The expansion coefficients are

$$c_1 = -2I'(2) = +0.33135119533744693185077637092564532, \quad (6)$$

$$c_2 = +2I''(2) = +0.28178934541054037839833656666846379. \quad (7)$$

The stability coefficient

$$\gamma = \frac{c_2}{I(2)} = 0.70283946799007245929257819650854315$$

(8)

This supersedes the published value $\gamma_{\text{published}} = 0.703051$, which differs by 0.03% due to rounding.

The entropy well

After the vacuum-tuning condition $c_1 = 0$,

$$\frac{S(\epsilon)}{S_0} = 1 - \gamma\epsilon^2 + O(\epsilon^4), \quad \epsilon \equiv D - 4. \quad (9)$$

The second derivative at $\epsilon = 0$ is

$$\left. \frac{d^2 S}{d\epsilon^2} \right|_{\epsilon=0} = -2\gamma = -1.405678935980144918585156393018 \dots < 0. \quad (10)$$

Thus $\epsilon = 0$ ($D = 4$) is a strict local maximum of the vacuum entropy.

Remark 1. The DME entropy well is the entire geometric input to ENTROPIA Core. Everything else in this paper is either algebraic manipulation of the well, or the standard non-equilibrium machinery needed to turn a static maximum into a dynamical attractor.

3 The Lyapunov functional

Definition 2 (Lyapunov functional). Define the *Lyapunov functional* as the normalized entropy deficit:

$$L[\epsilon] \equiv 1 - \frac{S(\epsilon)}{S_0}. \quad (11)$$

Using the DME expansion,

$$L[\epsilon] = \gamma\epsilon^2 + O(\epsilon^4). \quad (12)$$

Proposition 3. *The functional $L[\epsilon]$ satisfies:*

1. $L[0] = 0$.
2. $L[\epsilon] > 0$ for $\epsilon \neq 0$ (in a neighborhood of $\epsilon = 0$).
3. L is even in ϵ .

Proof. (1) follows from $S(0)/S_0 = 1$. (2) follows from $\gamma > 0$ and $\epsilon^2 > 0$. (3) follows from the Z_2 symmetry of the DME expansion, which contains only even powers of ϵ . $\square$

Thus L is a valid Lyapunov functional: non-negative, zero at the equilibrium, positive away from it, and even under the natural reflection symmetry of the entropy expansion.

4 EDF relaxation

The Lyapunov functional L by itself does not give a flow. To obtain a dynamical statement we import the Entropic Damping Force (EDF) result [2].

Setup

EDF models the relaxation of a visible-sector collective coordinate σ coupled weakly to a large hidden conformal bath. The microscopic action is

$$S = S_{\text{visible}} + S_{\text{hidden}} + S_{\text{portal}}, \quad (13)$$

with the visible sector containing a stiffness k near the infrared fixed point, the hidden sector consisting of N quantum scalars, and the portal providing a weak coupling.

Influence functional and friction

Integrating out the hidden bath yields an influence functional

$$\mathcal{F}[\sigma] = \exp\left(-\frac{1}{2}\sigma K\sigma + \dots\right). \quad (14)$$

For a dense spectrum with Ohmic spectral density $J(\omega) = \gamma_f\omega/2\pi$, short correlations produce a Markovian local friction. The effective equation of motion is

$$m\sigma''(\ell) + \gamma_f\sigma'(\ell) + k\sigma(\ell) = \xi(\ell), \quad (15)$$

where ℓ is the coarse-graining parameter, γ_f is the friction coefficient, k is the visible-sector stiffness, and ξ is stochastic noise satisfying the fluctuation–dissipation relation

$$\langle \xi(\ell)\xi(\ell') \rangle = 2\gamma_f T_b \delta(\ell - \ell'). \quad (16)$$

Overdamped limit

In the overdamped regime $\gamma_f \gg m$, the inertial term is negligible and the dynamics reduce to

$$\sigma'(\ell) = -\frac{k}{\gamma_f}\sigma(\ell). \quad (17)$$

EDF proves (§2.3, §3.2 of [2]) that linear damping is the unique form compatible with monotonic entropy production and absence of oscillations. Quadratic or higher-order corrections produce residual power-law or logarithmic tails that are observationally excluded.

Where EDF stops

EDF derives the *form* of the relaxation: linear, with rate $\beta = k/\gamma_f$. It does not identify σ with the DME dimensional deviation ϵ , and it does not fix the numerical value of k from DME. Those are the two steps that close ENTROPIA-Core.

5 The thermodynamic identification

We now close the gap between EDF and DME.

Identification 4 (Collective coordinate matching). The EDF collective coordinate σ and the DME dimensional deviation ϵ describe the same relaxation mode. They are related by

$$\sigma = c_\sigma \epsilon + O(\epsilon^2) \quad (18)$$

for some nonzero constant c_σ . Since both are small deviations from the same equilibrium, the linear term dominates, and the proportionality constant c_σ can be absorbed into the normalization of ϵ .

Identification 5 (Stiffness from entropy curvature). The EDF visible sector stiffness k is the curvature of the free energy $F = -TS$ at the DME equilibrium:

$$k \equiv \left. \frac{d^2 F}{d\epsilon^2} \right|_{\epsilon=0} = -T \left. \frac{d^2 S}{d\epsilon^2} \right|_{\epsilon=0} = 2TS_0\gamma. \quad (19)$$

Remark 6. The relation $F = -TS$ is the standard Legendre transformation between entropy and free energy. The identification $k = 2TS_0\gamma$ is therefore not a framework-specific postulate; it is the statement that the DME entropy well is the free-energy well of the EDF collective mode, up to the temperature factor T .

Assembling the gradient flow

Combining the two identifications with the EDF overdamped result:

$$\sigma'(\ell) = -\frac{k}{\gamma_f}\sigma(\ell). \quad (20)$$

Substituting $\sigma = c_\sigma \epsilon$ and $k = 2TS_0\gamma$:

$$\epsilon'(\ell) = -\frac{2TS_0\gamma}{\gamma_f}\epsilon(\ell). \quad (21)$$

Define the coarse graining parameter λ by the change of variables

$$\lambda \equiv \frac{2TS_0}{\gamma_f}\ell, \quad (22)$$

which absorbs the bath-specific ratio T/γ_f and the normalization S_0 into the definition of λ . Then

$$\boxed{\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon} \quad (23)$$

This is the ENTROPIA Core gradient flow. It is derived, not assumed.

Proposition 7 (Constraint on the hidden bath). *The identification $k = 2TS_0\gamma$ combined with the EDF overdamped rate $\beta = k/\gamma_f$ gives*

$$\frac{T_b}{\gamma_f} = \frac{\beta}{2S_0\gamma}. \quad (24)$$

For the EDF representative range $\beta \in [2, 10]$ and $S_0 = 1$, $\gamma = 0.703$,

$$T_b \approx (1.4 \text{ to } 7.1) \gamma_f. \quad (25)$$

The hidden bath is therefore predicted to be at a temperature of order its friction coefficient, which is consistent with EDF's own cold-bath assumption $T_b \ll \Lambda$ (EDF §4.2). The identification does not over-constrain the theory; it relates the bath temperature to the bath friction. The ratio T_b/γ_f is the only bath-specific quantity that enters ENTROPIA-Core, and it does not appear in the geometric sector.

Proof. EDF derives $\beta = k/\gamma_f$ in the overdamped limit. Substituting the identification $k = 2T_b S_0 \gamma$ and solving for T_b/γ_f gives the stated relation. The numerical range follows from EDF's representative $\beta \in [2, 10]$. $\square$

Remark 8. The relation $T_b \sim \gamma_f$ is a prediction, not a constraint. If a microscopic model of the hidden bath fixes T_b and γ_f independently, the relation becomes a consistency check. If the relation fails, the identification $k = 2TS_0\gamma$ fails, and the EDF stiffness must be sourced by a different mechanism.

Remark 9 (Separation of geometric and bath scales). The definition of λ involves the ratio T_b/γ_f , which is a property of the hidden bath, not of the DME geometry. The *shape* of the relaxation in λ is therefore fully determined by DME's γ ; the *physical* time scale is set by the bath. This separation is what allows ENTROPIA-Core to be a theorem of DME without introducing a new parameter in the geometric sector.

6 The ENTROPIA-Core system

We now collect the results.

Field

$$\epsilon(\lambda) = D(\lambda) - 4. \quad (26)$$

Lyapunov functional

$$L[\epsilon] = \gamma\epsilon^2 + O(\epsilon^4). \quad (27)$$

Gradient flow

$$\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon. \quad (28)$$

Lyapunov decrease

Compute

$$\frac{\partial L}{\partial \epsilon} = 2\gamma\epsilon + O(\epsilon^3). \quad (29)$$

Multiplying by $d\epsilon/d\lambda = -2\gamma\epsilon$:

$$\boxed{\frac{dL}{d\lambda} = -4\gamma^2\epsilon^2 + O(\epsilon^4) \leq 0} \quad (30)$$

The Lyapunov functional decreases monotonically along the flow. The equilibrium at $\epsilon = 0$ is globally attracting (in a neighborhood where the expansion is valid).

Remark 10 (Noise under the scale change). The fluctuation dissipation relation $\langle \xi(\ell)\xi(\ell') \rangle = 2\gamma_f T_b \delta(\ell - \ell')$ is written in the EDF coarse-graining variable ℓ . Under the change of variables $\lambda = (2T_b S_0 / \gamma_f)\ell$, the noise transforms as

$$\langle \xi(\lambda)\xi(\lambda') \rangle = 2\gamma_f T_b \left(\frac{2T_b S_0}{\gamma_f} \right)^{-1} \delta(\lambda - \lambda') = \frac{\gamma_f^2}{S_0} \delta(\lambda - \lambda'). \quad (31)$$

The noise amplitude in λ units is therefore fixed by γ_f and S_0 alone, with no residual T_b dependence. This is consistent with the fact that the mean trajectory $\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$ is independent of the bath temperature. Fluctuations about the mean are subleading for $T_b \ll \Lambda$ (EDF §4.2) and do not affect the ENTROPIA-Core result.

7 Master conditions

The Master conditions M1 and M2, previously treated as independent assumptions in the ENTROPIA-Core literature, now follow as consequences.

M1: Stationarity of effective information

Definition 11. The effective information $I_{\text{eff}}(\epsilon)$ is defined to be proportional to the entropy:

$$I_{\text{eff}}(\epsilon) \propto \frac{S(\epsilon)}{S_0} = 1 - \gamma\epsilon^2 + O(\epsilon^4). \quad (32)$$

Proposition 12. *M1 holds: $\partial_\lambda I_{\text{eff}}|_{\epsilon=0} = 0$.*

Proof. Compute

$$\frac{\partial I_{\text{eff}}}{\partial \epsilon} = -2\gamma\epsilon + O(\epsilon^3). \quad (33)$$

At $\epsilon = 0$, this vanishes. Since $\epsilon = 0$ corresponds to $\lambda = 0$ in the flow, M1 holds. $\square$

M2: Monotonic export

Definition 13. The exported relative entropy is defined as the integral of the Lyapunov decrease:

$$D_{\text{export}}(\lambda) = \int_0^\lambda \left(-\frac{dL}{d\lambda'} \right) d\lambda' = \gamma\epsilon_0^2 - \gamma\epsilon(\lambda)^2 + O(\epsilon^4). \quad (34)$$

Proposition 14. *M2 holds: $\partial_\lambda D_{\text{export}} \geq 0$ along the flow.*

Proof. Compute

$$\frac{\partial D_{\text{export}}}{\partial \lambda} = -\frac{\partial L}{\partial \lambda} = 4\gamma^2\epsilon^2 + O(\epsilon^4) \geq 0. \quad (35)$$

For any $\epsilon \neq 0$, this is strictly positive. At $\epsilon = 0$, it vanishes. $\square$

Remark 15. Both M1 and M2 are now consequences of the DME entropy well plus the definitions of I_{eff} and D_{export} . They are not independent assumptions.

8 Explicit solution and relaxation time

The gradient flow

$$\frac{d\epsilon}{d\lambda} = -2\gamma\epsilon \quad (36)$$

has the elementary solution

$$\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda} \quad (37)$$

where $\epsilon_0 = \epsilon(0)$ is the initial perturbation. Perturbations decay exponentially.

The relaxation time in λ units is

$$\tau_{\text{relax}} = \frac{1}{2\gamma} = \frac{1}{1.405678935980144918585156393018\dots} = 0.711400\dots \quad (38)$$

In physical units, the relaxation time is $\tau_{\text{relax}} \cdot (\gamma_f/2T_b S_0)$, where the second factor is a property of the hidden bath and is not fixed by DME alone. The *shape* of the decay is fully determined by γ .

9 Assumption audit

Table 1: Complete audit of ENTROPIA-Core. The gradient flow and the master conditions are derived; the only framework-specific input is the matching of the EDF and DME collective coordinates.

Item	Status	Source
$D = 4$	Input	FDLM/WIN
$\gamma = 0.702839467990\dots$	Derived	DME
$S/S_0 = 1 - \gamma\epsilon^2$	Derived	DME
Lyapunov $L = \gamma\epsilon^2$	Derived	Algebraic (DME)
EDF friction γ_f from Ohmic bath	Derived	EDF §2–3
Overdamped linear damping	Derived	EDF §3.2
Identification $\sigma = c_\sigma\epsilon$	Identified	This paper §5
Identification $k = 2T_b S_0 \gamma$	Identified	This paper §5
Gradient flow $d\epsilon/d\lambda = -2\gamma\epsilon$	Derived	EDF + identification
Lyapunov decrease $dL/d\lambda = -4\gamma^2\epsilon^2$	Derived	Algebraic
$M1: \partial_\lambda I_{\text{eff}} = 0$ at $\epsilon = 0$	Derived	Definition
$M2: \partial_\lambda D_{\text{export}} = 4\gamma^2\epsilon^2 \geq 0$	Derived	Definition
$\epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}$	Derived	Solution of flow
$\tau_{\text{relax}} = 1/(2\gamma) = 0.711400\dots$	Derived	Solution of flow
$T_b/\gamma_f = \beta/(2S_0\gamma)$	Derived	Prop. 7

The two identifications are the standard Legendre relationship between entropy and free energy, and the matching of two descriptions of the same relaxation mode. Neither is a framework-specific postulate. Neither introduces a free parameter into the geometric sector. The bath constraint of Proposition 7 is a prediction that can be tested against a microscopic model of the hidden bath.

10 Relation to DME and ENTROPIA-EFT

We emphasize that ENTROPIA-Core is a module, not a complete theory. It establishes the dynamical stability of the DME dimensional equilibrium. It does not attempt to derive the

effective field theory of the vacuum, the equation of state, the Einstein equations, or the cosmological observables. Those are the subject of ENTROPIA-EFT, and they are logically downstream of the present result.

Table 2: DME, ENTROPIA-Core, and ENTROPIA-EFT.

Layer	Provides	Status
DME	Static equilibrium: $S(\epsilon)/S_0 = 1 - \gamma\epsilon^2$, γ exact	Complete
ENTROPIA-Core	Dynamical stability: Lyapunov, gradient flow, master conditions M1, M2	Complete (this paper)
ENTROPIA-EFT	Effective field theory: $A1$ – $A4$, w formula, Einstein equations, Fisher metric, backreaction, H_0 boost	Partially open

What ENTROPIA-Core adds to DME

DME states that $D = 4$ is a strict maximum of the vacuum entropy. ENTROPIA-Core states that $D = 4$ is a dynamical attractor of the coarse-graining flow. The two together establish that the DME equilibrium is both statically preferred and dynamically stable.

What ENTROPIA-Core does not address

ENTROPIA-Core is a statement about the stability of the DME dimensional equilibrium. It does not derive:

- the w formula from the strip integral (an EFT problem),
- $A1$ (Proca) from the torus (an EFT problem),
- the Einstein equations from $A1$ – $A4$ (an EFT problem),
- the backreaction tensor $\Delta_{\mu\nu}$ (an EFT problem),
- the H_0 boost (an EFT/cosmology problem).

Those are the subjects of the ENTROPIA-EFT paper.

11 Conclusion

ENTROPIA-Core is derived from DME plus the EDF open-system relaxation. The chain is:

$$\begin{aligned}
\text{DME entropy well} &\rightarrow L[\epsilon] = \gamma\epsilon^2, \\
\text{EDF Ohmic bath} &\rightarrow \sigma' = -(k/\gamma_f)\sigma, \\
\text{Identifications} &\rightarrow d\epsilon/d\lambda = -2\gamma\epsilon, \\
&\rightarrow dL/d\lambda = -4\gamma^2\epsilon^2 \leq 0, \\
&\rightarrow \epsilon(\lambda) = \epsilon_0 e^{-2\gamma\lambda}, \\
&\rightarrow \tau_{\text{relax}} = 1/(2\gamma) = 0.711400\dots
\end{aligned}$$

The identification also constrains the hidden-bath ratio T_b/γ_f to be of order unity, consistent with EDF’s cold-bath assumption. No framework-specific postulate is required beyond the two identifications of §5. All numerical content is inherited from DME’s γ and is exact.

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A DME expansion coefficients

For reference, the DME coefficients used in this paper, evaluated to 50-digit precision:

$$I(2) = 0.40092988263220389628003875385983806, \quad (39)$$

$$c_1 = 0.33135119533744693185077637092564532, \quad (40)$$

$$c_2 = 0.28178934541054037839833656666846379, \quad (41)$$

$$\gamma = 0.70283946799007245929257819650854315. \quad (42)$$

The published DME value $\gamma_{\text{published}} = 0.703051$ differs from the exact value by 0.03% and is superseded.

B EDF derivation summary

For completeness, we summarize the EDF chain used in §4:

1. **Axioms.** Open-system RG; Gaussian hidden sector; Ohmic spectral density $J(\omega) \propto \omega$; stable infrared fixed point.
2. **Influence functional.** Gaussian integration of the hidden bath yields a quadratic influence functional.
3. **Local friction.** Ohmic spectrum produces a Markovian local friction kernel $\gamma_f \sigma'(\ell)$.
4. **Effective equation.** $m\sigma'' + \gamma_f \sigma' + k\sigma = \xi$.
5. **Overdamped limit.** For $\gamma_f \gg m$: $\sigma'(\ell) = -(k/\gamma_f)\sigma(\ell)$.
6. **Uniqueness.** Linear damping is the unique form compatible with monotonic entropy production and absence of oscillations.

The EDF paper establishes that the dissipation arises purely in theory space: the full microscopic system remains unitary, and irreversibility appears only after tracing over the hidden bath.

C Consistency between the noise rescaling and Proposition 7

The noise rescaling of Remark 10 and the bath constraint of Proposition 7 are consistent. Combining them:

$$\frac{\langle \xi(\lambda) \xi(\lambda') \rangle}{\delta(\lambda - \lambda')} = \frac{\gamma_f^2}{S_0} = \frac{k^2}{S_0 \beta^2} = \frac{(2T_b S_0 \gamma)^2}{S_0 \beta^2}. \quad (43)$$

The noise amplitude in λ units is therefore a function of T_b and γ only through the ratio $T_b/\gamma_f = \beta/(2S_0\gamma)$. In the physical regime $T_b \ll \Lambda$ and $\beta \in [2, 10]$, the noise amplitude is bounded. No singularity appears as $T_b \rightarrow 0$. This confirms that the ENTROPIA-Core result is robust against the details of the hidden bath.